

Quiz 1

Student Name: _____

Student ID: _____

A factory operates three production lines: L_1 , L_2 , and L_3 .

Their shares of total output and defect probabilities are given below:

Line	Fraction of total output	Defect probability
L_1	40 %	3 %
L_2	35 %	5 %
L_3	25 %	8 %

Instructions

- Write all **relevant probability formulas** (law of total probability, Bayes' theorem, independence definition) **before substituting numbers**.
- Use clear mathematical notation and show all steps.
- **Define event labels explicitly** before calculations. For example:
 - L_1 : "Product is from Line 1"
 - L_2 : "Product is from Line 2"
 - L_3 : "Product is from Line 3"
 - D : "Product is defective"

Questions

1. (4 pts) Find the probability that a randomly selected product is defective ($P(D)$).
2. (4 pts) If a product is defective, what is the probability it was produced by L_3 ($P(L_3 | D)$)?
3. (4 pts) Line L_3 installs a new inspection system that detects **75%** of its defective items before shipment, while L_1 and L_2 detect none. Detected defective items will not be sent to customers. If a defective product **reaches a customer**, what is the probability it was produced by L_3 ?
4. (4 pts) Let L_3 : "Product is from Line 3" and D : "Product is defective."
Are these events **mutually exclusive**? Are they **independent**? Explain briefly.
5. (4 pts) If all production lines had the **same defect rate**, would L_3 and D become independent? Why or why not?